

Rational inequalities

Never cross-multiply. Move everything to one side, combine into one fraction, and let the sign chart do the work.

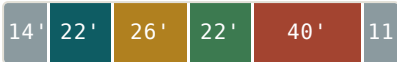
LESSON 32–34

3 × 40 MIN

ALGEBRA 10, §2.3

P1 · 1.2 (EXTENSION)

LESSON CLOCK



The forbidden step 14 min

One fraction, compared with zero 22 min

The sign chart 26 min

Repeated factors 22 min

Practice 40 min

Homework 11 min

LEARNING OBJECTIVES

- Explain why multiplying an inequality by a variable expression is forbidden.
- Reduce a rational inequality to a single fraction compared with zero.
- Build and read a sign chart, including the excluded points.
- Write the answer in interval notation.

TEXTBOOK

National pp. 114–128

Cambridge Pure Mathematics 1, pp. 6–11

Workbook P1 Exercise 1C

01 Explanation

The one thing you may not do

NEVER MULTIPLY AN INEQUALITY BY $x - 3$

You do not know its sign. If it is negative the inequality reverses; if it is zero it collapses. $\frac{1}{x} > 1$ does **not** give $1 > x$ — try $x = -1$: $-1 > 1$ is false, but $1 > -1$ is true.

Multiplying by a **positive number** is safe. Multiplying by an expression whose sign you do not know is not, and there is no way to repair it afterwards.

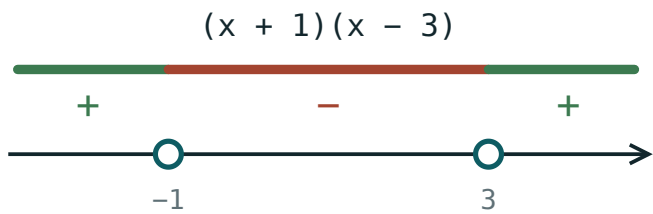
The method

FOUR STEPS, ALWAYS

1 Move everything to one side, so the other side is 0. **2** Combine into a single fraction and factorise top and bottom. **3** Mark every zero of the numerator and of the denominator on a number line. **4** Test the sign in each interval and read the answer off.

$$\frac{1}{x} > 1 \Rightarrow \frac{1}{x} - 1 > 0 \Rightarrow \frac{1-x}{x} > 0$$

Critical values 0 and 1. Testing -1, 0.5, 2 gives -, +, -, so the answer is $0 < x < 1$.



negative between the roots, positive outside

The critical values cut the line into intervals. One test point settles each.

Open or closed?

CRITICAL VALUE FROM	INEQUALITY	ENDPOINT
numerator	$\leq$ or $\geq$	included — closed circle
numerator	$<$ or $>$	excluded — open circle
denominator	any	always excluded

A DENOMINATOR ZERO IS NEVER INCLUDED

Even with $\geq$. The expression has no value there at all, so it cannot satisfy anything.

Repeated factors

The sign alternates as you cross a simple zero. It does **not** alternate at a repeated factor of even order — the graph touches the axis and turns back.

$$\frac{(x - 1)^2(x + 2)}{x - 3} \geq 0$$

X	< -2	-2	$(-2,1)$	1	$(1,3)$	3	> 3
sign	+	0	-	0	-	x	+

Answer: $x \leq -2$, or $x = 1$, or $x > 3$. The isolated point 1 is part of the solution because the numerator is zero there and $\geq$ allows it.

A QUICK RULE

Start at the far right, where every factor is positive, and count the sign changes leftwards. A factor of odd power changes the sign; a factor of even power does not.

02 Worked examples

EXAMPLE 1 Solve $\frac{x-2}{x+3} < 0$.

- ① Already one fraction against zero.
- ② Critical values 2 and -3 .
- ③ Test $-4, 0, 3$: $+, -, +$.
- ④ Both ends open.

ANSWER

$$-3 < x < 2$$

EXAMPLE 2 Solve $\frac{2}{x-1} \leq 3$.

① $\frac{2}{x-1} - 3 \leq 0$

② $\frac{2-3(x-1)}{x-1} = \frac{5-3x}{x-1} \leq 0$

③ Critical $x = \frac{5}{3}$ (closed), $x = 1$ (open).

④ Test 0, 1.2, 2: -, +, -.

ANSWER

$$x < 1 \text{ or } x \geq \frac{5}{3}$$

EXAMPLE 3 Solve $\frac{(x+1)(x-4)}{x^2} > 0$.

① Critical $-1, 0, 4$.
 $x = 0$ is a double root of the denominator.

② Signs: +, -, -, + across the four intervals.
 x^2 never changes sign.

③ Ends all open.

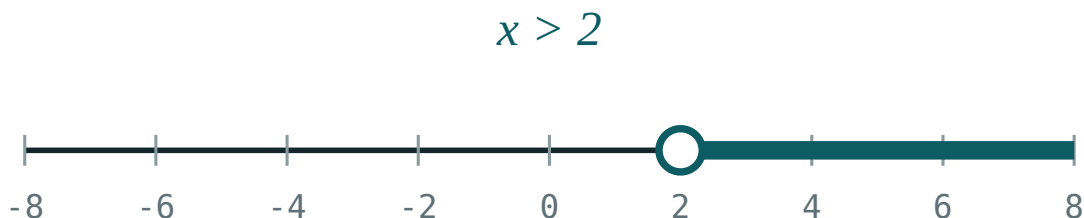
ANSWER

$$x < -1 \text{ or } x > 4$$

03 Interactive model

Draw the number line first, then fill in the signs interval by interval.

Reading a sign chart

inequality $x > 2$ boundary **2** (open) $x = 0$ works? **no** $x = 6$ works? **yes****04** Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Rational inequality	Ratsional tengsizlik	Рациональное неравенство
Sign chart (interval method)	Oraliqlar usuli	Метод интервалов
Critical value	Kritik qiymat	Критическое значение
Excluded point	Chetlatilgan nuqta	Выколота́я точка
Closed circle	To'ldirilgan nuqta	Закрашенная точка
Open circle	Bo'sh nuqta	Выколота́я точка
Interval notation	Oraliq belgilanishi	Интервальная запись
Union of intervals	Oraliqlar birlashmasi	Объединение промежутков
Sign of a factor	Ko'paytuvchi ishorasi	Знак множителя
Multiplicity	Karralilik	Кратность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

1 $x > 1$ gives:

A zero of the denominator is:

Crossing a factor of even power the sign:

$x - 2$ $x + 3 < 0$ gives:

$$x < -3$$

$$-3 < x < 2$$

$$x > 2$$

$$x < 2$$

The first step of a rational inequality is:

cross-multiply

get zero on one side

square it

take reciprocals

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\frac{1}{x} > 0$

ANSWER $x > 0$

2. Solve $\frac{1}{x} < 0$

ANSWER $x < 0$

3. Solve $\frac{x}{x-1} > 0$

ANSWER $x < 0$ or $x > 1$

4. Solve $\frac{x-3}{x+1} < 0$

ANSWER $-1 < x < 3$

5. Solve $(x-1)(x+2) > 0$

ANSWER $x < -2$ or $x > 1$

6. Is $x = 2$ allowed in $\frac{1}{x-2} \geq 0$?

ANSWER no

7. Solve $\frac{2}{x} \geq 0$

ANSWER $x > 0$

Medium · the standard question

7 PROBLEMS

1. Solve $\frac{1}{x} > 1$

ANSWER $0 < x < 1$

2. Solve $\frac{2}{x-1} \leq 3$

ANSWER $x < 1$ or $x \geq \frac{5}{3}$

3. Solve $\frac{x+2}{x-3} \geq 0$

ANSWER $x \leq -2$ or $x > 3$

4. Solve $\frac{x}{x^2-4} > 0$

ANSWER $-2 < x < 0$ or $x > 2$

5. Solve $\frac{(x+1)(x-4)}{x^2} > 0$

ANSWER $x < -1$ or $x > 4$

6. Solve $\frac{3}{x+2} < 1$

ANSWER $x < -2$ or $x > 1$

7. Solve $\frac{x-1}{x+1} \leq 2$

ANSWER $x < -1$ or $x \geq -3 \rightarrow x \leq -3$ or $x > -1$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\frac{(x-1)^2(x+2)}{x-3} \geq 0$

ANSWER $x \leq -2$, $x = 1$, or $x > 3$

2. Solve $\frac{x^2 - 5x + 6}{x^2 - 4} \leq 0$

ANSWER $-2 < x \leq 2$ excluding $2 \rightarrow -2 < x < 2$ or $x = 3$

3. Solve $\frac{1}{x-1} > \frac{1}{x+1}$

ANSWER $x > 1$ or $-1 < x \rightarrow -1 < x < 1$ fails; $x > 1$ or $x < -1$

4. Solve $\frac{x}{x-2} \geq \frac{3}{x-2}$

ANSWER $x > 2$

5. Solve $\frac{x^3 - x}{x+3} < 0$

ANSWER $x < -3$ or $-1 < x < 0$ or $x > 1$ — check signs

6. Find all k with $\frac{x^2 + k}{x} > 0$ for every $x > 0$

ANSWER $k \geq 0$

7. Explain why $\frac{(x-2)^2}{x-2}$ and $x-2$ give different solution sets

ANSWER The first excludes $x = 2$

07 Homework

Homework — 6 tasks

Every answer needs the number line drawn, with open and closed circles marked.

1. Solve $\frac{x-5}{x+2} < 0$.

2. Solve $\frac{3}{x-2} \geq 1$.

3. Solve $\frac{x}{x^2-9} \leq 0$.

4. Solve $\frac{(x-2)^2(x+1)}{x-5} \geq 0$.

5. Explain in three sentences why $\frac{1}{x} > 1$ is not $1 > x$.

6. Solve $\frac{2}{x+1} < \frac{1}{x}$.